

# Omid Arefian

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in Omid Arefian

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## Education

2024 – Present    📖 **M.Sc. in Mechanical Engineering (Mechatronics & Robotics), Politecnico di Milano**  
Cumulative GPA: 24.88/30

2019 – 2024    📖 **B.Sc. in Mechanical Engineering, Shahid Beheshti University, Tehran**  
GPA: 17.77/20

## Employment History

March 2026 – Present    📖 **Thesis Internship, Suncubes (Full-time), Milan, Italy**  
Working on drone pose estimation for autonomous navigation and sensing applications.

- Designing and implementing a Multiplicative Extended Kalman Filter (MEKF) for drone attitude and pose estimation.
- Developing the estimation framework in Python using PyMAVLink, PX4 simulation, and QGroundControl.
- Testing and validating the filter using both PX4 simulation data and real experimental drone measurement data.
- Developing sensor-fusion algorithms using IMU, GPS, barometer.
- Investigating robustification techniques to improve MEKF reliability under noisy measurements, disturbances, and sensor uncertainty.
- Implementing a Federated Kalman Filter architecture to fuse laser-based measurements with the MEKF pose-estimation pipeline.

Jun 2023 – Feb 2024    📖 **Research Assistant, Shahid Beheshti University (Part-time), Tehran, Iran**  
Research assistant to Professor Roghayeh Gavagsaz-Ghoachani in the Energy Engineering Laboratory.

- Leveraged experience to guide undergraduate students, enhancing their academic progression.
- Prepared six conference posters for international and domestic conferences.
- Collaborated with master's and doctoral students to foster knowledge sharing.
- Developed a comprehensive understanding of article publishing and research workflows.
- Gained expertise in photovoltaic panels technologies.

## Projects

- Fall 2025
- **Advanced Nonlinear Control Design, Politecnico di Milano**
    - Modeled a vehicle in a 2D environment and computed optimal obstacle-avoiding trajectories using direct and indirect methods.
    - Designed Model Predictive Control (MPC) and Linear Quadratic Regulator (LQR) controllers to solve the optimal control problem.
  - **Path-Following Controller Design for TurtleBot3 in Gazebo using MATLAB and ROS2, Politecnico di Milano**
    - Integrated ROS2 with MATLAB under Ubuntu, enabling seamless simulation-to-hardware bridging.
    - Designed and validated linear and nonlinear controllers, including PID and MPC, for precise path tracking in cluttered environments.
- Summer 2025
- **Custom Reinforcement-Learning Environment & Agent, Politecnico di Milano**
    - Developed a PPO-based reinforcement learning simulation to coordinate multiple boxes, preventing collisions and arranging them in a straight line.
    - Designed the agent to compute and apply optimal linear and angular velocity commands in a physics-driven environment.
    - Demonstrated consistent collision avoidance and reliable alignment across varied initial conditions.
- Spring 2025
- **System Identification of a Steel Plate, Politecnico di Milano**
    - Conducted spectral FRF testing on a plate, modelling on Ansys to find the modal parameters.
    - Implemented  $H_1$  and  $H_2$  estimators with FFT-based normalization and windowing.
    - Developed a model-reduction workflow to approximate the plate's multi-degree-of-freedom (MDOF) dynamics by equivalent single-degree-of-freedom (SDOF) systems.
    - Evaluated coherence and resonance features to select the most reliable FRF approach.
  - **Experimental Modal Analysis of Light-Rail Wheel, Politecnico di Milano**
    - Configured a free-free wheel test using a dynamometric impact hammer and 12 piezo-electric accelerometers.
    - Computed FRFs and coherence in MATLAB to extract axial modes' natural frequencies, damping ratios, and mode shapes.
    - Employed MATLAB-based fitting methods to refine FRF models against experimental measurements.

## Skills

- Languages
- English (C1), Italian (A2).
- Software & Tools
- Python, MATLAB, PyMAVLink, PX4, QGroundControl, ROS2, Gazebo, ANSYS, SolidWorks.

## Research Interests

- Optimal Control and Model Predictive Control
- UAV Guidance, Navigation, and Control

## Research Interests (continued)

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- State Estimation and Sensor Fusion
- Motion Planning and Path Planning

## Awards and Honors

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- 2023: Invited to present a peer-reviewed paper at an international conference in Thailand.
- 2019–2023: Full Bachelor's Scholarship at Shahid Beheshti University, awarded for outstanding academic merit.
- 2019: Top 1 % in National University Entrance Exam, ranking among the very best participants.